Anatomy of a Hong–Ou–Mandel Dip of Quantum-Dot Photons: Lifetime, Emission-Time Jitter and Coherence from a Single Delay Scan

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Abstract

We analyse a Hong–Ou–Mandel (HOM) delay scan of two single photons emitted by a semiconductor quantum dot 12.1 ns apart. Starting from the two-time coherence of a dephased two-level emitter we derive a closed-form dip shape: an exponential of the radiative lifetime convolved with the distribution of the photons' emission-time difference. Among seven candidate shapes this model (and its coherent variant) is the only one that describes the 130 data points at the noise level (residual rms 1.7×10^{-3}, random residuals), whereas the pure exponential, Lorentzian and Gaussian dips are rejected by $\Delta\mathrm{AICc} > 300$. The fit yields a zero-delay visibility $V_0 = 0.832 \pm 0.001_{\mathrm{stat}} \pm 0.004_{\mathrm{model}}$, a jitter-free interference amplitude $V_{\mathrm{int}} = 0.920 \pm 0.002$, a radiative lifetime $T_1 = 130.8 \pm 0.7 \pm 1.9$ ps, an rms emission-time jitter of 11.9 ± 0.3 ps per photon, a coherence time $T_2 \geq 218$ ps and a homogeneous linewidth $\lesssim 6.0$ μeV, 1.2 times the Fourier limit. Every number is accompanied by an independent check, and all scripts are provided.

1. Introduction

Two-photon interference on a beam splitter (**?**) is the standard benchmark of single-photon sources for linear-optical quantum technologies (**??**). When two identical photons meet on a balanced beam splitter they always leave through the same port, so the probability of a coincidence between the two outputs vanishes; any distinguishability in polarisation, spectrum or arrival time restores coincidences (**?**). Scanning the relative delay τ of the photons therefore traces a *dip* whose depth measures the photon indistinguishability and whose *shape* is fixed by the photons' temporal modes.

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Quantum-dot (QD) sources are usually characterised only by the depth of this dip (**??**). The shape, however, contains much more: the radiative lifetime, the statistics of the emission time, and — in combination with the depth — the coherence time and linewidth of the emitter. Extracting this information requires (i) a physically motivated model of the dip and (ii) a careful statistical procedure that shows which features of the model are actually supported by the data.

In this work we analyse one such delay scan. Our contributions are:

- A closed-form model of the delay-integrated HOM dip of a dephased two-level emitter with random emission time (Section 3), and an independent numerical verification of it.
- A quantitative comparison of seven dip shapes with information criteria, residual diagnostics, cross-validation and simulation-based power analysis (Section 4).
- A full set of source parameters with statistical and model uncertainties, and an explicit list of the assumptions each one depends on (Sections 5 and 6).

2. Data

The data set (`data/data.txt`) contains $n = 130$ values of the normalised coincidence level C between the two outputs of a beam splitter, recorded as a function of the position x of one of the beam-splitter inputs, $x = 0$–13.545 cm in steps of 0.105 cm (Figure 1). The two photons are emitted by the same QD 12.1 ns apart (consecutive excitation pulses) and are brought to temporal overlap by a delay line; x fine-tunes the remaining delay. No error bars, count rates or auxiliary measurements ($g^{(2)}(0)$, beam-splitter ratio, lifetime) are given.

Delay scale. Moving the input by Δx changes the optical path by Δx, i.e. $\tau = n_g \Delta x / c$ with the group index of air $n_g = 1.00028$ (1 cm $\leftrightarrow$ 33.36 ps).

Noise level. A model-free estimate of white noise from second differences, $\hat{\sigma}^2 = \langle (y_{i+1} - 2y_i + y_{i-1})^2 \rangle / 6$, gives $\hat{\sigma} = 1.45 \times 10^{-3}$; this is the benchmark a correct model must reach.

3. Model of the dip

HOM with mixed photons. For photons a and b with normalised two-time coherences $G_{a,b}(t,t') = \langle \hat{a}^\dagger(t)\hat{a}(t') \rangle$ incident on a beam splitter with reflectivity R and transmissivity $T = 1 - R$, the delay-integrated coincidence probability is (**???**)

$$P_c(\tau) = R^2 + T^2 - 2RT\, J(\tau), \qquad (1)$$
$$J(\tau) = \iint G_a(t,t')\, G_b(t'-\tau, t-\tau)\, \mathrm{d}t\, \mathrm{d}t'.$$

J is the overlap $\mathrm{Tr}\, \rho_a \rho_b$ of the (in general mixed) single-photon states; it is bilinear in the states, so classical mixtures (e.g. random emission times) enter as averages of G.

Dephased two-level emitter. A QD excited at t_0 and decaying with rate $\Gamma = 1/T_1$ under pure dephasing γ^* emits a photon with $G(t,t') = \Gamma e^{-\Gamma(t+t'-2t_0)/2}\, e^{-\gamma^*|t-t'|}$ for $t, t' > t_0$. Inserting it into Equation (1) gives (Section A)

$$J(\tau) = \frac{T_2}{2T_1}\, e^{-|\tau|/T_1}, \qquad \frac{1}{T_2} = \frac{1}{2T_1} + \gamma^*. \qquad (2)$$

Dephasing lowers the depth but not the shape; a slowly wandering central frequency (constant within one photon) likewise only lowers the depth, since $|\langle \psi_a | \psi_b \rangle|^2 = e^{-|\tau|/T_1}/(1 + \delta^2 T_1^2)$ for a detuning δ. Detector timing jitter is irrelevant for delay-integrated coincidences.

Emission-time jitter. Under pulsed excitation the photon does not start at a fixed time: relaxation from higher states, phonon-assisted processes or a finite pulse length randomise t_0 (**??**). Averaging Equation (2) over the distribution of $\Delta t_0 = t_{0a} - t_{0b}$ convolves the exponential with that distribution. For Gaussian jitter of rms σ_1 per photon (Δt_0 Gaussian with $s = \sqrt{2}\sigma_1$), with $L = cT_1$ and $u = |x - x_0|$,

$$K_{\mathrm{EG}}(u) = \tfrac{1}{2} e^{s^2/2L^2} \Big[e^{-u/L}\, \mathrm{erfc}\, \tfrac{s/L - u/s}{\sqrt{2}} + e^{u/L}\, \mathrm{erfc}\, \tfrac{s/L + u/s}{\sqrt{2}} \Big], \qquad (3)$$

which rounds the cusp of the exponential and reduces the zero-delay overlap to $K_{\mathrm{EG}}(0) = e^{s^2/2L^2}\, \mathrm{erfc}(s/\sqrt{2}L) < 1$. For exponential feeding with time constant l_r, Δt_0 is Laplace-distributed and $K_{\mathrm{EL}}(u) = \frac{b}{b^2 - a^2}(be^{-au} - ae^{-bu})$, $a = 1/L$, $b = 1/l_r$.

Fit function. Collecting all depth-reducing effects in one amplitude, the model is

$$C(x) = A\, \big[1 - V_{\mathrm{int}}\, K_{\mathrm{EG}}(x - x_0;\, L, s) \big], \qquad (4)$$

Table 1. Model comparison ($n = 130$). σ: residual rms; runs: observed/expected number of sign runs; p_{runs}: two-sided runs-test p-value; DW: Durbin–Watson; CV: 10-fold cross-validated rms prediction error for random / contiguous folds (in 10^{-3}).

model	k	$10^3\sigma$	ΔAICc	runs	DW	CV
exp∗Gauss	5	1.69	0.1	59/66	1.65	1.70/1.99
coherent	5	1.69	0	61/66	1.65	1.70/1.96
exp∗Laplace	5	1.75	8.6	58/65	1.56	1.78/4.02
asymmetric	6	1.70	2.0	57/66	1.66	–
exponential	4	5.41	301	27/66	0.30	5.69/12.5
Lorentzian	4	8.04	404	13/66	0.10	8.26/31.5
Gaussian	4	16.7	594	5/66	0.04	17.4/53.3

with five parameters: the level A for distinguishable photons, the jitter-free amplitude V_{int}, the zero-delay position x_0, the lifetime length L and the jitter width s. The observable visibility at zero delay is $V_0 = V_{\mathrm{int}} K_{\mathrm{EG}}(0) = 1 - C(x_0)/A$. For a balanced beam splitter and $g^{(2)}(0) = 0$, $V_{\mathrm{int}} = T_2/2T_1$.

Alternatives. We also fit (i) a *coherent* variant in which the jitter is replaced by a pure wave packet $\psi = g * e^{-t/2T_1}$ shaped by a Gaussian pulse g (then $J = [K_{\mathrm{EG}}(u; 2L, s)/K_{\mathrm{EG}}(0; 2L, s)]^2$ and the amplitude equals V_0 directly); (ii) the pure exponential Equation (2); (iii) Laplace jitter; (iv) Gaussian and (v) Lorentzian dips (expected for Gaussian or exponential spectra, e.g. filtered parametric sources (**?**)); (vi) an asymmetric variant with separate L on the two sides of x_0, as a symmetry test.

4. Fitting and model selection

All models are fitted by unweighted least squares (Levenberg–Marquardt). We compare them with the corrected Akaike criterion AICc and BIC (**?**), and test the residuals with the Wald–Wolfowitz runs test, the Durbin–Watson statistic (DW; 2 for uncorrelated residuals) and the Shapiro–Wilk normality test. Uncertainties are 68% intervals from 3000 residual-bootstrap and 1000 moving-block-bootstrap replicas (block length 8) (**?**).

Results. Table 1 and Figure 2 summarise the comparison. The Gaussian and Lorentzian shapes leave oscillating residuals of $\pm$20–50 $\times 10^{-3}$ and are rejected: the dip has exponential wings, as expected for spontaneous emission (Lorentzian spectrum). The pure exponential describes the wings but not the bottom of the dip, which is rounded over $\approx \pm 0.5$ cm (Figure 3); the residuals are strongly correlated (DW $= 0.30$, $p_{\mathrm{runs}} = 6 \times 10^{-12}$). Adding Gaussian emission-time jitter reduces the residual rms by a factor of 3.2, to within 17% of the model-free noise estimate, and makes the residuals compatible with white Gaussian noise ($p_{\mathrm{runs}} = 0.23$, Shapiro–Wilk $p = 0.31$; the mild DW

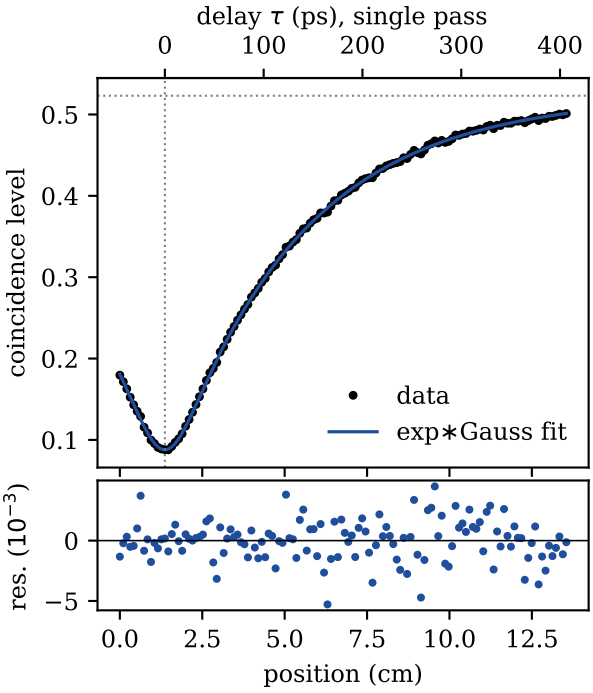


Figure 1. Coincidence level versus input position (dots) and the fit of the exponential∗Gaussian model, Equation (4) (line). Dotted lines: A and x_0. Top axis: delay for $p = 1$. Bottom: residuals, rms 1.69×10^{-3}.

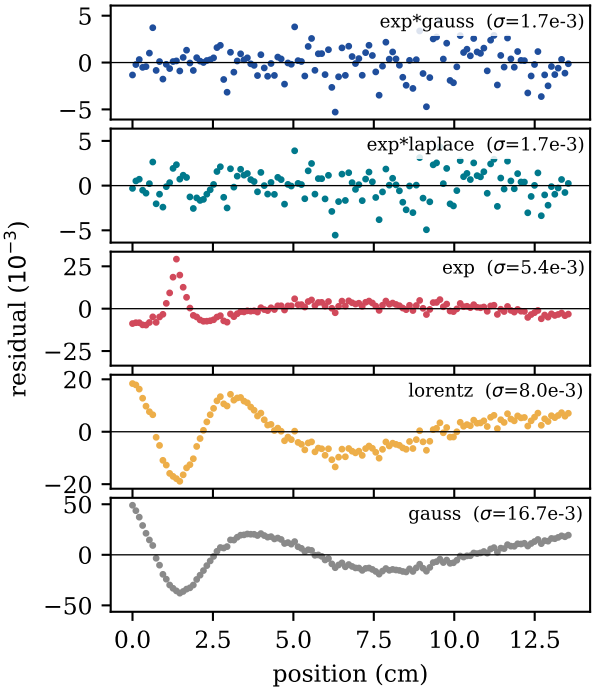


Figure 2. Residuals of five dip models (note the different vertical scales). Only the models with emission-time jitter give structureless residuals.

$= 1.65$ is covered by the block bootstrap).

Shape of the jitter. Gaussian jitter is preferred over exponential (Laplace) jitter by $\Delta\text{AICc} = 8.5$, and the Laplace model fails markedly in contiguous cross-validation (4.0 vs 2.0×10^{-3}), because it predicts a too-deep minimum ($C_{\min} = 0.0857$ vs. 0.0878). To check that this preference is not a statistical fluctuation we simulated 300 data sets from the best Laplace model with the observed noise: the Gaussian model was preferred in only 7 of them, and never by as much as observed ($p < 0.003$).

Coherent vs. incoherent. The coherent pulse-shaped wave packet and the incoherent Gaussian jitter give numerically indistinguishable shapes ($\Delta\text{AICc} = 0.1$): as long as the jitter is much shorter than T_1 both reduce to an exponential with a rounded tip. The data therefore fix the width of the rounding, but not whether it is coherent. This ambiguity matters only for the interpretation of V_{int}, not for V_0, which is identical in both models (Table 2).

Symmetry. Fitting separate decay lengths left and right of the minimum gives $L_1 = 4.00 \pm 0.14$ cm and $L_2 = 3.92 \pm 0.02$ cm, consistent with a symmetric dip; the extra parameter is not supported ($\Delta\text{AICc} = +2.0$).

5. Physical parameters

Table 2 lists the results. We discuss them in order of decreasing robustness.

Zero-delay visibility. $V_0 = 0.832$ is the most robust number: it is identical in the incoherent and coherent models, changes by at most 0.004 for Laplace jitter, and a model-light estimate (parabola through the seven lowest points, $C_{\min} = 0.08788$, with A from a tail-only fit) gives 0.831. For a balanced beam splitter, perfect mode matching and $g^{(2)}(0) = 0$ it equals the two-photon indistinguishability M; all experimental imperfections lower the visibility, so $M \geq V_0$. With the correction of **?**, $M = (V_0 + g^{(2)})/(1 - g^{(2)})$, one would obtain $M = 0.851$, 0.869 and 0.929 for $g^{(2)}(0) = 0.01$, 0.02 and 0.05.

Lifetime. The exponential wings fix $L = 3.92$ cm, i.e. $T_1 = 131$ ps, a Purcell-shortened lifetime typical of QDs in micropillar cavities (**??**). The value is stable to 0.5% on all subsets of the data (Section B) and changes by 1.5% between the acceptable models. The single-photon wave packet is $cT_1 = 3.9$ cm long.

Emission-time jitter. The rounding of the tip corresponds to an rms jitter of Δt_0 of 16.9 ± 0.4 ps, i.e. $\sigma_1 = 11.9 \pm 0.3$ ps per photon (Laplace model: feeding time 14 ps;

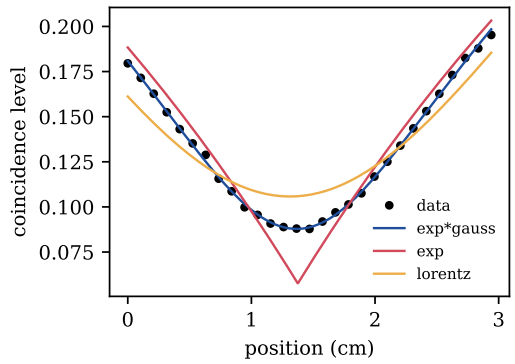


Figure 3. Bottom of the dip. The pure exponential has a cusp at x_0 that the data do not show; the Lorentzian is too flat. The rounding is reproduced by an emission-time jitter of $\sigma_1 \approx 12$ ps.

coherent model: pulse intensity rms 8.6 ps). Such values are typical of excitation pulses of several ps or of phonon-assisted/relaxation-based excitation (**??**). The jitter alone limits the overlap to $K_{\rm EG}(0) = 0.905$: if it is incoherent, it accounts for about half of the total visibility loss $1 - V_0$.

Coherence time and linewidth. Because $V_{\rm int} \le T_2/2T_1$ (equality for a balanced beam splitter and $g^{(2)} = 0$), the dip yields a *lower* bound on T_2: $T_2 \ge 2V_{\rm int}T_1 = 241$ ps if the jitter is incoherent, or $T_2 \ge 2V_0T_1 = 218$ ps in the most conservative case. The homogeneous linewidth is correspondingly at most 1.46 GHz (6.0 μeV), only 1.1–1.2 times the Fourier limit of 1.22 GHz (5.0 μeV); the pure-dephasing time is $T_2^* \gtrsim 1.3$ ns. Since the two photons are separated by 12.1 ns, spectral wandering slower than this does not enter; the numbers characterise dephasing on the $\lesssim$10 ns scale.

Normalisation. The asymptote $A = 0.523$ exceeds the value $1/2$ expected for distinguishable photons on a balanced beam splitter, and forcing $A = 1/2$ is excluded (the last points already exceed 0.50; even a bi-exponential kernel then gives $\sigma = 6.3 \times 10^{-3}$). If C were an absolute probability, Equation (1) would imply $R^2 + T^2 = A$, i.e. $R \approx 0.39$, and a maximal visibility $2RT/(R^2 + T^2) = 0.912$. This is compatible with V_0 (giving a corrected $M \approx 0.913$), but is violated by $V_{\rm int} = 0.920$ at 4.2σ. Thus either the data are normalised to an arbitrary reference, or the jitter is (at least partly) coherent; the present data cannot decide. Also, if $V_{\rm int}$ were the true M, positivity of $1 - M$ would bound $g^{(2)}(0) \le (1 - V_{\rm int})/2 = 0.040$.

Zero delay. The overlap position $x_0 = 1.366$ cm marks where the delay line compensates the 12.1 ns separation exactly (an unfolded path of $c \cdot 12.1$ ns $= 3.63$ m in air). It is determined to ± 40 μm, i.e. ± 0.13 ps.

Table 2. Parameters of the exp∗Gauss model and derived quantities. Errors: residual bootstrap (block bootstrap agrees within 10%); where given, the second error is the spread among the three acceptable models (exp∗Gauss, coherent, exp∗Laplace). V_0: zero-delay visibility; $V_{\rm int}$: jitter-free amplitude; $K_{\rm EG}(0)$: jitter-limited overlap; σ_1: rms emission-time jitter per photon; $T_2^* = 1/\gamma^*$; $\Delta\nu = 1/\pi T_2$; $1/2\pi T_1$: Fourier limit; cT_2: coherence length. Pairs "a / b" are obtained from V_0 / $V_{\rm int}$.

quantity	value
A (distinguishable level)	0.5231 ± 0.0006
x_0 (cm)	$1.366 \pm 0.004 \pm 0.004$
V_0	$0.8321 \pm 0.0012 \pm 0.004$
$V_{\rm int}$	0.9196 ± 0.0015
$K_{\rm EG}(0)$	0.905 ± 0.002
L (cm)	3.921 ± 0.020
s (cm)	0.506 ± 0.011
dip FWHM (cm)	6.28 ± 0.02
T_1 (ps)	$130.8 \pm 0.7 \pm 1.9$
jitter σ_1 (ps)	11.9 ± 0.3
dip FWHM (ps)	209.7 ± 0.7
T_2 (ps)	218 / 241
T_2^* (ns)	1.30 / 2.99
$\Delta\nu$ (GHz)	1.46 / 1.32
$\hbar\Delta\omega$ (μeV)	6.05 / 5.47
$1/2\pi T_1$ (GHz)	1.217
cT_2 (cm)	6.5 / 7.2

6. Verification

Each claim above was checked independently of the fit that produced it (scripts in `src/`, logs in `results/`):

1. **Analytic kernel.** $J(\tau)$ was computed by brute-force double integration of G_aG_b on a time grid (`verify_model.py`), for $T_1 = 130$ ps, $T_2^* = 3$ ns, $\sigma_1 = 12$ ps, without using Equations (2) and (3). The relative deviation from $\frac{T_2}{2T_1}K_{\rm EG}$ is 2.5×10^{-3} at a 1 ps grid and 1.3×10^{-3} at 0.5 ps, i.e. it vanishes linearly with the step. The coherent-model formula agrees to $< 10^{-4}$. The closed form Equation (3) also agrees with direct quadrature of the convolution to better than 10^{-15}.

2. **Global minimum.** Differential evolution over wide bounds reproduces the Levenberg–Marquardt parameters to $< 4 \times 10^{-8}$.

3. **Error bars.** In 400 injection–recovery simulations the fit is unbiased ($|\text{bias}| < 0.1\sigma$) and the covariance errors have 65–69% coverage; residual- and block-bootstrap errors agree within 10%. A fit weighted with Poisson-like errors $\propto \sqrt{C}$ changes $V_{\rm int}$, L and s by less than 1σ.

4. **Model selection.** AICc, BIC and cross-validation give the same ranking; the Gauss-vs-Laplace preference was tested by simulation (Section 4).

5. **Robustness.** Fits to subsets (only $x \leq 9$ cm, $x \geq 0.5$ cm, without the minimum, even or odd points) agree within 1σ (Section B).

6. **Visibility.** Model-light estimate of V_0 (0.831) agrees with the model (0.832).

7. Limitations

The beam-splitter ratio, $g^{(2)}(0)$, polarisation and spatial mode overlap are not known, so V_0 and V_{int} only bound M and T_2 from below. The model assumes identical photons from the two pulses (same T_1 and spectrum). The noise model is empirical: without raw counts we cannot test Poisson statistics directly. An independent time-resolved lifetime measurement would test T_1; a $g^{(2)}$ measurement and a scan with orthogonal polarisations (which gives A directly) would convert the bounds on M and T_2 into values.

8. Conclusion

A single HOM delay scan of QD photons contains far more than a visibility. The dip is exponential in its wings, with a rounded tip; a two-level emitter with dephasing and ≈ 12 ps Gaussian emission-time jitter reproduces all 130 points at the noise level. From the fit we obtain the lifetime (131 ps), the jitter, the zero-delay visibility (0.832), a jitter-free amplitude (0.920), and lower bounds on the coherence time (≥ 218 ps) and on how close the photons are to the Fourier limit (within a factor 1.2). The main systematic uncertainty is the delay scale, which can be removed by a single auxiliary measurement.

Software and Data

The analysis code is in `src/` (models, fits, verification) and its output in `results/`; the raw data in `data/data.txt` were not modified. `python src/fit_hom.py` reproduces all tables and figures.

Impact Statement

This paper presents work whose goal is to advance the characterisation of single-photon sources for quantum technologies. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

A. Derivation of the kernel

With $G(t,t') = \Gamma e^{-\Gamma(t+t')/2} e^{-\gamma^*|t-t'|}\theta(t)\theta(t')$ for photon a and the same function shifted by $\tau > 0$ for photon b,

$$J(\tau) = \Gamma^2 \iint_{t,t'>\tau} e^{-\Gamma(t+t')/2}\, e^{-\Gamma(t+t'-2\tau)/2}\, e^{-2\gamma^*|t-t'|}\,\mathrm{d}t\,\mathrm{d}t' = \Gamma^2 e^{-\Gamma\tau} \iint_{u,u'>0} e^{-\Gamma(u+u')} e^{-2\gamma^*|u-u'|}\,\mathrm{d}u\,\mathrm{d}u'$$
$$= 2\Gamma^2 e^{-\Gamma\tau} \int_0^\infty e^{-2\Gamma u}\,\mathrm{d}u \int_0^\infty e^{-(\Gamma+2\gamma^*)v}\,\mathrm{d}v = \frac{\Gamma}{\Gamma+2\gamma^*}\, e^{-\Gamma\tau} = \frac{T_2}{2T_1}\, e^{-|\tau|/T_1},$$

using $u = t-\tau$, $v = u'-u$ and $1/T_2 = \Gamma/2 + \gamma^*$; $\tau < 0$ follows by symmetry. With random start times, $G \to \langle G_{t_0}\rangle$; since J is bilinear and depends only on $\tau + t_{0a} - t_{0b}$, it becomes the convolution of Equation (2) with the probability density of $\Delta t_0 = t_{0a} - t_{0b}$. For $\Delta t_0 \sim \mathcal{N}(0, s^2)$, $\int e^{-|u-w|/L}\,\mathcal{N}(w;0,s^2)\,\mathrm{d}w$ evaluates to Equation (3) by splitting the integral at $w = u$ and completing the square. For a coherent pulse, $\psi = g * \phi$ with $\phi(t) = \sqrt{\Gamma} e^{-\Gamma t/2}\theta(t)$ and the overlap amplitude is the autocorrelation $(g \star g) * (\phi \star \phi)$; for a Gaussian $|g|^2$ of rms σ_p, $g \star g$ is Gaussian with $s = 2\sigma_p$ and $\phi \star \phi = e^{-|\tau|/2T_1}$, which gives the coherent model of Section 3.

B. Parameter stability and bootstrap distributions

Table 3. exp∗Gauss fits on subsets of the data (errors from the covariance matrix).

subset	A	V_{int}	x_0 (cm)	L (cm)	s (cm)
all points	0.5231(6)	0.9196(15)	1.3662(41)	3.921(20)	0.506(11)
$x \le 9$ cm	0.5236(19)	0.9187(18)	1.3655(38)	3.938(41)	0.501(12)
$x \ge 0.5$ cm	0.5231(7)	0.9186(19)	1.3700(58)	3.922(21)	0.498(14)
without $\lvert x - x_0\rvert < 0.3$ cm	0.5230(7)	0.9197(16)	1.3662(42)	3.919(21)	0.509(16)
even points	0.5228(8)	0.9196(20)	1.3687(51)	3.911(26)	0.508(15)
odd points	0.5233(10)	0.9196(24)	1.3635(65)	3.930(32)	0.504(18)

The correlation coefficients of the main fit are largest between A and L ($+0.91$), V_{int} and L (-0.79), and V_{int} and s ($+0.78$): a longer lifetime is partly traded against a higher asymptote, and a deeper intrinsic dip against a wider jitter. The bootstrap distributions (Figure 4) are unimodal and nearly Gaussian.

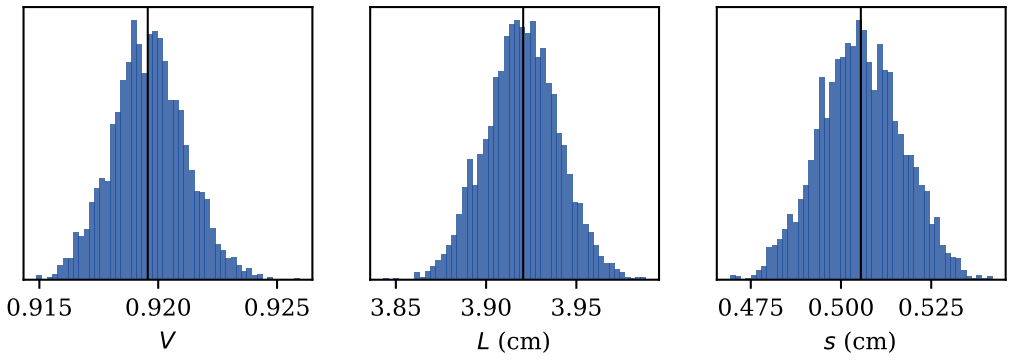


Figure 4. Residual-bootstrap distributions (3000 replicas) of V_{int}, L and s; vertical lines: best fit.